

Teaching Statement

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As a teacher and mentor, I strive to create approachable settings for a wide range of students to apply, internalize and deliberate on knowledge with sufficient context for application and space for creativity. To achieve this in a rapidly evolving world, I continuously create and adapt both digital and classroom resources that motivate students to engage in ① active learning and ② collaborative feedback ③ so they can build confidence and critical thinking to navigate a fast-evolving world. My background has prepared me to teach undergraduate courses in human-AI interaction, software engineering, data structures and algorithms as well as graduate-level courses covering special topics on the future of work and technology, user-centered design and computational social science.

Teaching

In graduate school, I was the sole TA for two iterations of Introduction to Human AI Interaction (HAII) – which ranges ~50-00 students, and Introduction to Software Construction – an invite-only course targeted to improve retention of those concurrently enrolled in a core CS major pre-requisite. In these roles I created syllabi, prepared and delivered new lectures, managed homeworks and designed course policies. During my undergraduate degree, I served as a teaching assistant for three courses, where I grew in a variety of TA (including lecturing) roles in a liberal arts setting. I continuously seek opportunities to hone my teaching and mentoring, such as by taking courses in [CS Pedagogy](#) (CMU), [Language Pedagogy](#) (Oberlin) and [Graduate Student Mentor Training](#).

Guiding Active and Applied Learning

Knowledge is not a copy of reality [...] To know an object is to act on it [...] to transform the object, and to understand the process of this transformation, and as a consequence to understand the way the object is constructed [3]

As researchers and knowledge producers, we are experts in creating and delivering accurate and up-to-date course material, yet student engagement with course content rely primarily on relevance and their perceived subjective value. To incentivize hands-on engagement with content-heavy lectures in HAII, I integrated structured pre-class readings and discussions to guide students in developing their own perspectives. At the end of each class session, I led collaborative in-class activities that encourage active application of lecture material through divergent and explorative thinking. As an example, students in the first iteration of HAII thoroughly immersed themselves in the Black Mirror Writers' Room¹ — an in-class activity to speculate on the harms of future technologies. But despite their investment reflecting on ethics and societal impacts of AI, I worried how this critical lens might foster a fear of advancements. Thus, when I [redesigned course materials](#), I included a complementary “*Light Mirror*” exercise where students develop second-episode plots – directing them to look beyond the problem space and practice devising and critiquing their own (alternative) solutions. To encourage in-depth engagement in future teaching, I will incorporate exercises (e.g. socratic debates) that require more critical and reflective thinking.

Collaborative, Creative and Constructive Peer Feedback

The “deep secret” of collaborative learning seems to lie in the cognitive processes through which humans progressively build a shared understanding [4]

In addition to course content, *how* students engage with material in the classroom also influences engagement and retention. One way method I use to encourage more creative ideas and approaches is to

¹ Based on [method introduced by Casey Fiesler](#)

structure collaborative spaces and activities where students can imitate, critique and build off of each others' understandings – such activities also imitate real-world development contexts more faithfully, where interdisciplinary team members carry varying and shifting objectives [5].

Collaboration is easier said than done. Even among small group environments such as in office hours, individual students might take unique approaches of their own, making it necessary to personalize teaching and mentorship with customized scaffolding. As an undergrad TA and office hours holder for Oberlin's introductory algorithm course, I observed how collaborative students (those who helped others reach the last piece of a problem/puzzle) received not only the satisfaction of successfully giving peer support, they also understood the challenge and embedded concepts better – because the act of solving from another's perspective helped them gain fuller scope and understanding. In addition to creating classroom opportunities for student collaboration, I also make sure to prepare each course I manage with digital spaces (*e.g.* Piazza, Slack) for asynchronous collaboration, so that students can exchange support without the frictions of physical or temporal distance. Finally, one technique that worked effectively for me as a tutor was to reflexively build up student knowledge by asking them critical but incremental questions, to connect their existing knowledge while probing for understanding of necessary course concepts. When developing future course materials that cover more abstract topics, I plan to design in-class activities that help students collaboratively construct knowledge in similar grounded and progressive ways.

Co-evolving with Students in an Ever-Changing World

To work toward a just world – a world where all have equal access to opportunity – means, as a start, opening up heart and mind to the perspectives of others. We must be able to hear each other and to respect and learn from what we hear [1]

As experts and deliverers of course content, it is our duty to regularly watch for and adapt to pivotal changes in the larger educational and research environments, so that our learning objectives align with evolving student needs. For instance, during the Spring iteration 2023 of HAI I sensed an intense, collective student fear of job displacement by the (then recent) LLM advances – conflicting with their rational goals to develop fluency in fundamental programming, as well as theoretical and practical algorithmic literacy – skills prioritized by the CS education I received. At the same time, these student concerns were genuine, rational and valid, with potential to impact their motivations to learn. Thus, in following lectures, I made sure to introduce contextualizing material about the past, present, and potential futures of computing that can help students navigate increasingly digital and AI-augmented worlds.

I highlighted historical parallels of the current Digital Revolution to the First Industrial Revolution: showing how displaced workers were absorbed by rising manufacturing demands, just as AI currently replaces clerical tasks but opens up more creative and knowledge-based jobs. Transitioning to the present, I added lectures (during [course redesign](#)) on (1) the evolution of AI support for writing — from early spell-check tools to more mature assistance and generative tools of today, as well as (2) approaches for detecting and mitigating AI failures. Towards the future, I introduced design methods (*e.g.* speed dating, storyboarding, co-design) to surface multiple possibilities. Joining institutional efforts to adapt to recent advances, I participated in the workshop *Adapting to Generative AI in the Classroom* at CMU's Eberly Center to consider new ways of assessment and material development. Moving forward, I am committed to situating students and ourselves within the material we deliver (from early on) to (1) improve perceptions of content relevance and (2) guide students toward more grounded and practical perspectives.

Mentoring

As a mentor, I guide students to develop interpersonal and research skills, explore interests, and provide resources and direction for projects over three stages. My 13 mentees (11 undergraduate, 2 masters) from diverse backgrounds (9 different institutions) imparted me with valuable, cross-disciplinary collaborations and perspectives. Two of these mentees are now pursuing PhDs of their own at NYU and UPenn.

Defined Beginnings: To start, I outline higher-level objectives and concrete weekly milestones to clearly communicate expectations in writing. This practice (associated with the [Independent Development Plan for qualitative research](#) I developed for the [Undergraduate Research Engagement Working Group](#)) began with my first batch of summer students in 2022. Beyond onboarding and short-term goals, I also include readings and resources to help mentees orient and contextualize their roles in prior research.

Feedback-based Execution: While short-term goals help deliver individual tasks, longer-term aims (e.g. project objectives, career directions/ambitions) require more iterative exploration and refinement to identify. To support *research skill* development, I incrementally provide feedback on how mentees synthesize, communicate, and build upon research findings via guided milestones (e.g. literature surveys, stand-ups, and retrospective syncs). *Interpersonally*, I strongly encourage mentees to protest project directions and technical struggles as early as possible in their research, so they can garner courage and normalize asking for help. In a recent game design project where I pushed mentees to challenge my ideas, one repeatedly raised immersion as a key objective – which eventually expanded not only my understanding on the persuasive potentials of gamified interventions, but also the overall project scope.

Guided Reflections & Future Directions: After their designated time with the project, 7 of 9 summer students end up continuing with us to wrap up final pieces and reflexively orient toward their long-term research interests. Even during the course of the project, I schedule mid-summer check-ins to assess and reevaluate mentees' expected contributions as well as lighthearted end-of-week check-ups about weekend plans (reverse ice-breakers) to remind them to transition from work to life. In addition to supporting my own students in finding their future directions, I also co-organized for CMU's [graduate application support program](#) to scalably match aspiring applicants with current computing PhD students to give feedback on statements. In 2024 alone, the program matched more than 100 external applicants with CMU mentors. As a faculty member, I hope to implement similar programs to bridge gaps between different levels of student communities, especially those from historically underrepresented backgrounds.

Reference

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- [4] Weiss, G., & Dillenbourg, P. (1999). What is 'multi' in multi-agent learning? In P. Dillenbourg (Ed.), *Collaborative Learning: Cognitive and Computational Approaches* (pp. 64–80).
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